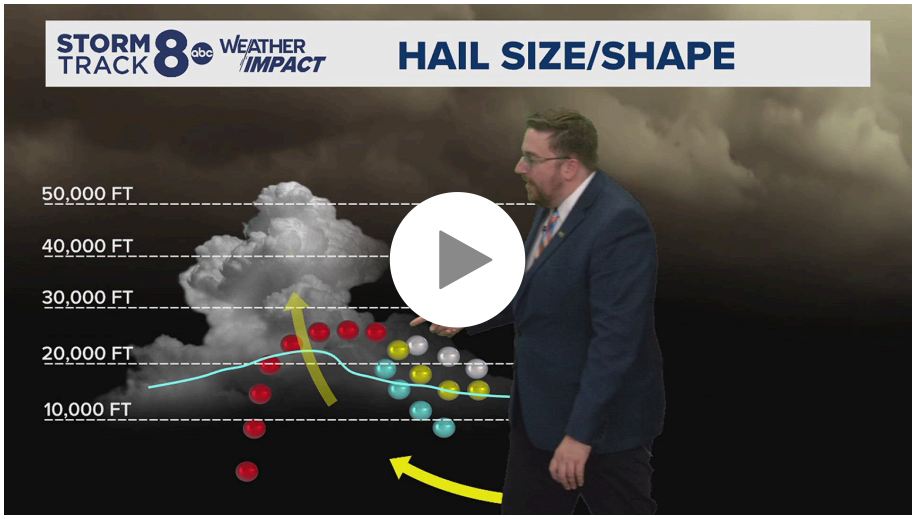
ASK ANDREW

From spheres to spikes: Decoding the wild shapes of hail

Meteorologist Andrew Stutzke explains why some hail is round while others have a more spikey appearance.



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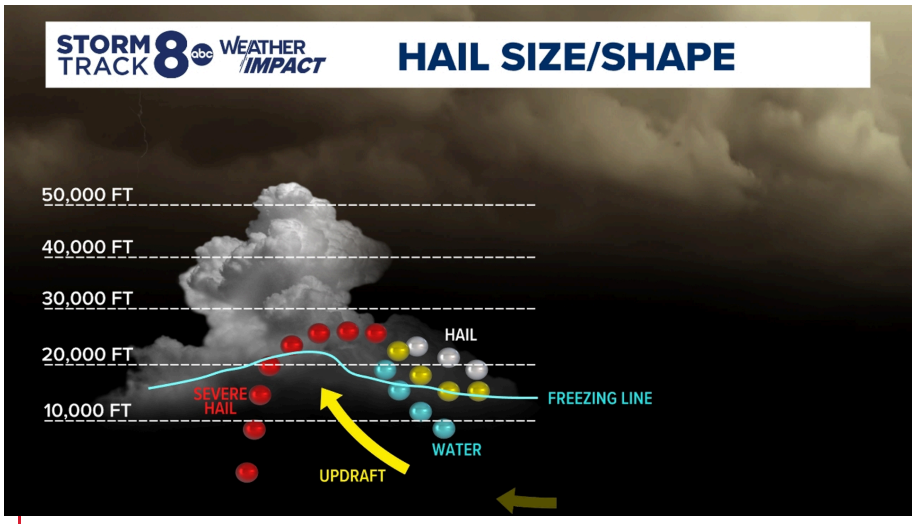
Author: Andrew Stutzke

MOLINE, Ill. — Recently, we talked about why hail varies in size depending on different wind speeds within severe thunderstorms. What about the shape, though? Diane from Rock Island recently asked that same question during an episode of Ask Andrew Live. Let's dig in!

Hail formation 101

Before we can get into the details regarding hail shape, we have to start with the basics first: How hail itself forms.

Hailstones start as water droplets that are carried high into a cumulonimbus cloud associated with a thunderstorm. These water droplets enter a layer of the atmosphere that is below freezing, moving up and down within the storm as they are tossed around by strong updraft and downdraft winds.



Credit: WQAD

Hail forms within storms when strong updrafts carry water droplets into the atmosphere, where they freeze and collide with supercooled water droplets,

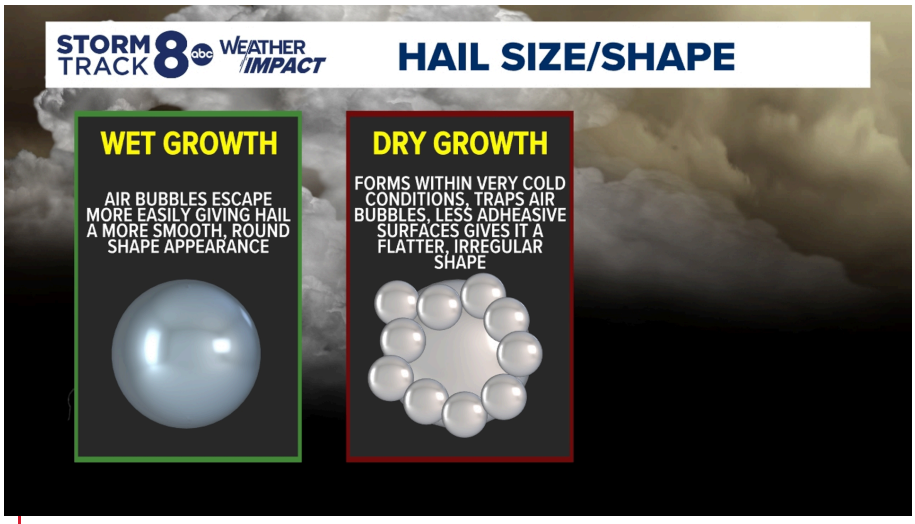
As the hailstones circulate, they encounter different temperature levels, allowing them to form layers of ice. This constant movement and collisions with supercooled water droplets and other ice particles allow them to grow. Eventually, gravity takes over and the hail becomes too heavy to be suspended within the cloud. It then falls to the ground.

Two growth processes

Hail shape is mostly determined by the growth process involved in forming the hail itself. There are two different processes: wet growth and dry growth

Wet Growth: Occurs when the hailstone collects liquid supercooled water, creating a clear layer of ice. These hailstones are rounder and smooth.

Dry Growth: This occurs when the hailstone collects ice crystals and creates a cloudy layer of ice.



Credit: WQAD

Different growth processes and environmental conditions will determine the size and shape of hail.

Conditions with thunderstorms are constantly changing, which causes the hailstone to go through periods of wet and dry growth, creating the layered appearance seen when hailstones are cut open. This layering also contributes to the varying shapes of hail.

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